
$$27.7) \quad \lambda = \frac{h}{p} \Rightarrow p = \frac{h}{\lambda}$$

$$E = \frac{p^2}{2m} = \left(\frac{h}{\lambda}\right)^2 \frac{1}{2m} = \left(\frac{6.63 \times 10^{-34}}{10^{-10}}\right)^2 \frac{1}{2 \times 9.11 \times 10^{-31}}$$

$$= 2.4 \times 10^{-17} \text{ J} = 150 \text{ eV.}$$

$$27.8) \quad \lambda = \frac{h}{p} = \frac{h}{mv} = \frac{6.63 \times 10^{-34}}{0.15 \times 30} = 1.47 \times 10^{-34} \text{ m}$$

$$27.12) \quad E = \frac{p^2}{2m}, \quad p = \frac{h}{\lambda}$$

$$\Rightarrow E = \frac{h^2}{2m\lambda^2} \Rightarrow \lambda^2 = \frac{h^2}{2mE}$$

$$\lambda = \frac{h}{\sqrt{2mE}} = \frac{6.63 \times 10^{-34}}{\sqrt{2 \times 16 \times 1.66 \times 10^{-27} \times 1.6 \times 10^{-19}}}$$

$$= 7.19 \times 10^{-12} \text{ m}$$
